

Cryptography and Network Security Lecture - 1



Personal Preference

- **Q:** Which of these would you choose for a day off?
- a) Sleeping in 🛌
 - b) Playing video games 🎮
 - c) Going for a walk 🚶♀️
 - d) Hanging out with friends 🧑🧑



Computer Security



- **Q:** Which of these is NOT a strong password?
 - a) \$Tr0ngPa\$\$w0rd!
 - b) 123456
 - c) CorrectHorseBatteryStaple



Computer Security

- **Q:** Which of these is NOT a strong password?
 - a) \$Tr0ngPa\$\$w0rd!
 - b) 123456
 - c) CorrectHorseBatteryStaple

- **A:** b) 123456



General Knowledge

- What is the name of malicious software that locks your files and demands payment?



General Knowledge

- What is the name of malicious software that locks your files and demands payment?
- **A:** Ransomware.



General Knowledge

➤ **Q:** What does “phishing” typically involve?



General Knowledge

- **Q:** What does “phishing” typically involve?
- **A:** Tricking users into giving up personal information by pretending to be a trusted source.



Mobile Security

- **Q:** What is one simple step you can take to secure your mobile device?



Mobile Security

- **Q:** What is one simple step you can take to secure your mobile device?
- **A:** Set a strong screen lock (PIN, password, or biometric).



Wireless Security

- Why should you avoid using public Wi-Fi without a VPN for sensitive activities?



Wireless Security

- Why should you avoid using public Wi-Fi without a VPN for sensitive activities?
- **A:** Because public Wi-Fi can allow attackers to intercept your data.



Cloud Security

- **Q:** True or False: It is your cloud provider's sole responsibility to secure your data in the cloud.



Cloud Security

- **Q:** True or False: It is your cloud provider's sole responsibility to secure your data in the cloud.
- **A:** False. Security in the cloud follows a **shared responsibility model** where the provider secures the infrastructure, and you secure your data and access.



Introduction

The art of war teaches us to rely not on the likelihood of the enemy's not coming, but on our own readiness to receive him; not on the chance of his not attacking, but rather on the fact that we have made our position unassailable.

—The Art of War, Sun Tzu





Agenda



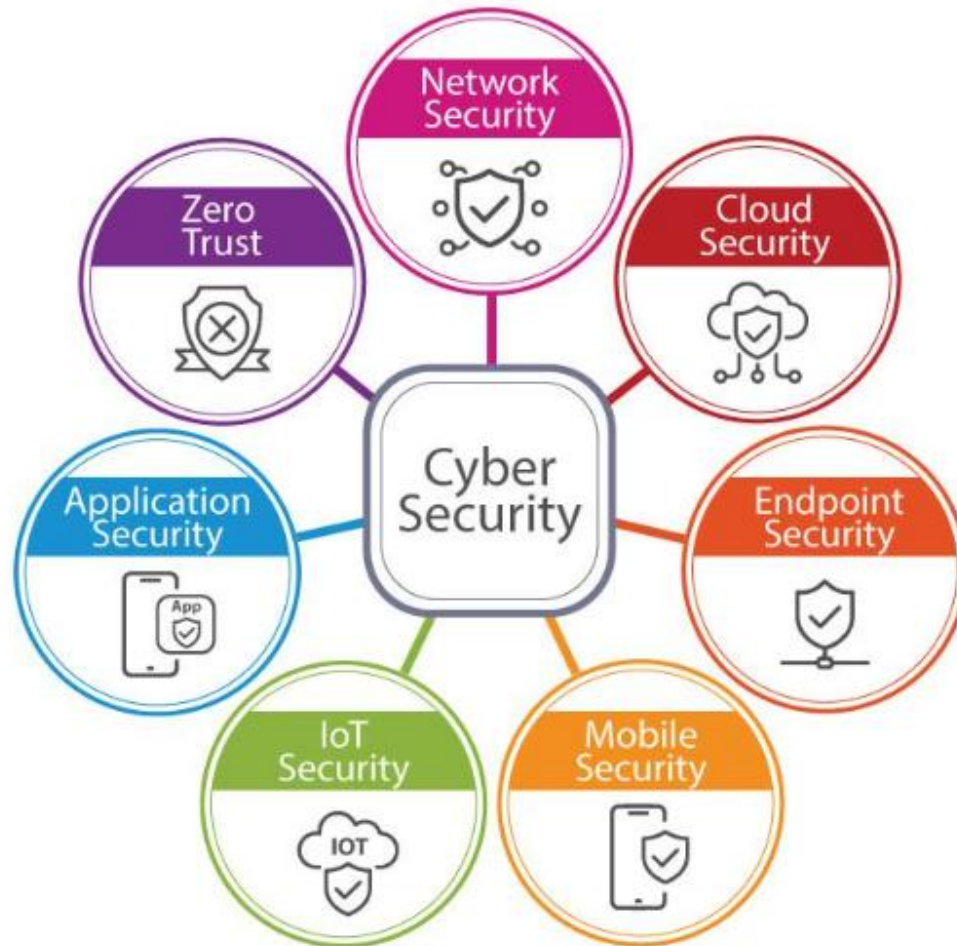
- Introduction to Computer Security & Cryptography
 - • Overview of Computer Security
 - • Importance of cybersecurity
 - • Threat landscape
 - • Basics of confidentiality, integrity, and availability (CIA Triad)
 - • Fundamentals of Cryptography
 - • History of cryptography
 - • Symmetric vs. asymmetric encryption • Common algorithms (AES, RSA)



Computer security

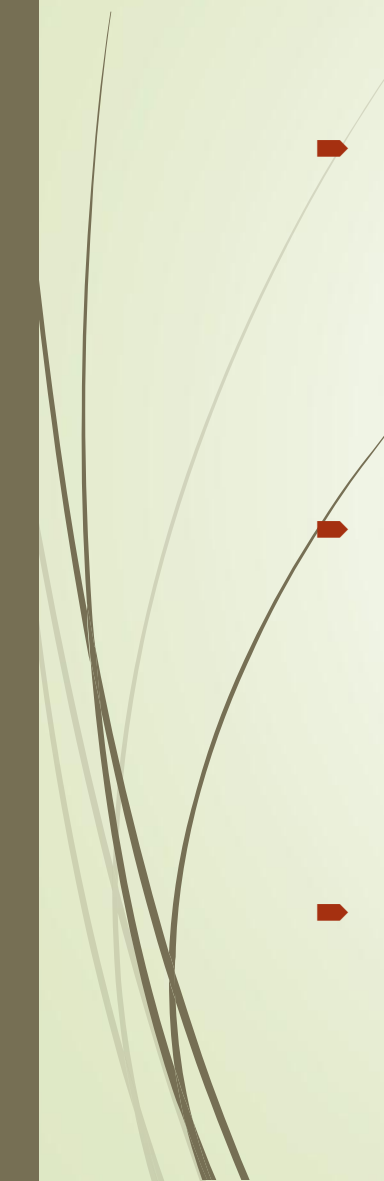
- Computer security refers to protecting and securing computers and their related data, networks, software, hardware from unauthorized access, misuse, theft, information loss, and other security issues. The Internet has made our lives easier and has provided us with lots of advantages but it has also put our system's security at risk of being infected by a virus, of being hacked, information theft, damage to the system, and much more.

Types of computer security





Types of computer security



➤ 1. Network Security:

- Protects computer networks from unauthorized access, misuse, or theft.
- Uses tools like firewalls, intrusion detection systems, and VPNs to monitor and control network traffic.
- Focuses on securing the infrastructure that connects devices and allows data to flow.

➤ 2. Application Security:

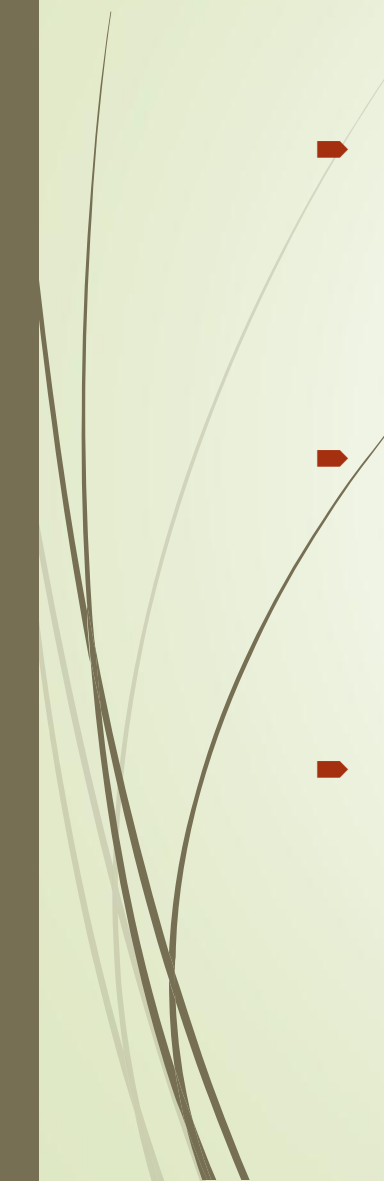
- Focuses on securing software and applications from cyber threats throughout their lifecycle.
- Involves secure coding practices, regular vulnerability assessments, and the use of web application firewalls.
- Aims to prevent attacks that exploit vulnerabilities in applications.

➤ 3. Information Security (Data Security):

- Ensures the confidentiality, integrity, and availability of sensitive data.
- Uses techniques like encryption, access controls, and data loss prevention to protect data at rest and in transit.



Types of computer security



➤ 4. Cloud Security:

- Protects data, applications, and infrastructure hosted in cloud environments.
- Cloud security is often a shared responsibility between the organization and the cloud service provider.

➤ 5. Endpoint Security:

- Secures individual devices like computers, laptops, and smartphones from cyber threats.
- Includes measures like antivirus software, endpoint detection and response (EDR) tools, and device management policies.

➤ 6. IoT Security:

- Focuses on securing interconnected devices and networks within the Internet of Things (IoT) ecosystem.
- IoT devices are often overlooked but can be a major entry point for attackers.



Types of computer security

➤ 7. Operational Security:

- Deals with the processes and decisions related to managing and protecting data assets.
- Focuses on how data is handled, stored, and accessed, ensuring it is done securely.

➤ 8. Human Layer Security:

- Recognizes that humans are often the weakest link in cybersecurity.
- Focuses on educating and training users to recognize and avoid phishing attacks, social engineering, and other threats.

➤ 9. AI Security:

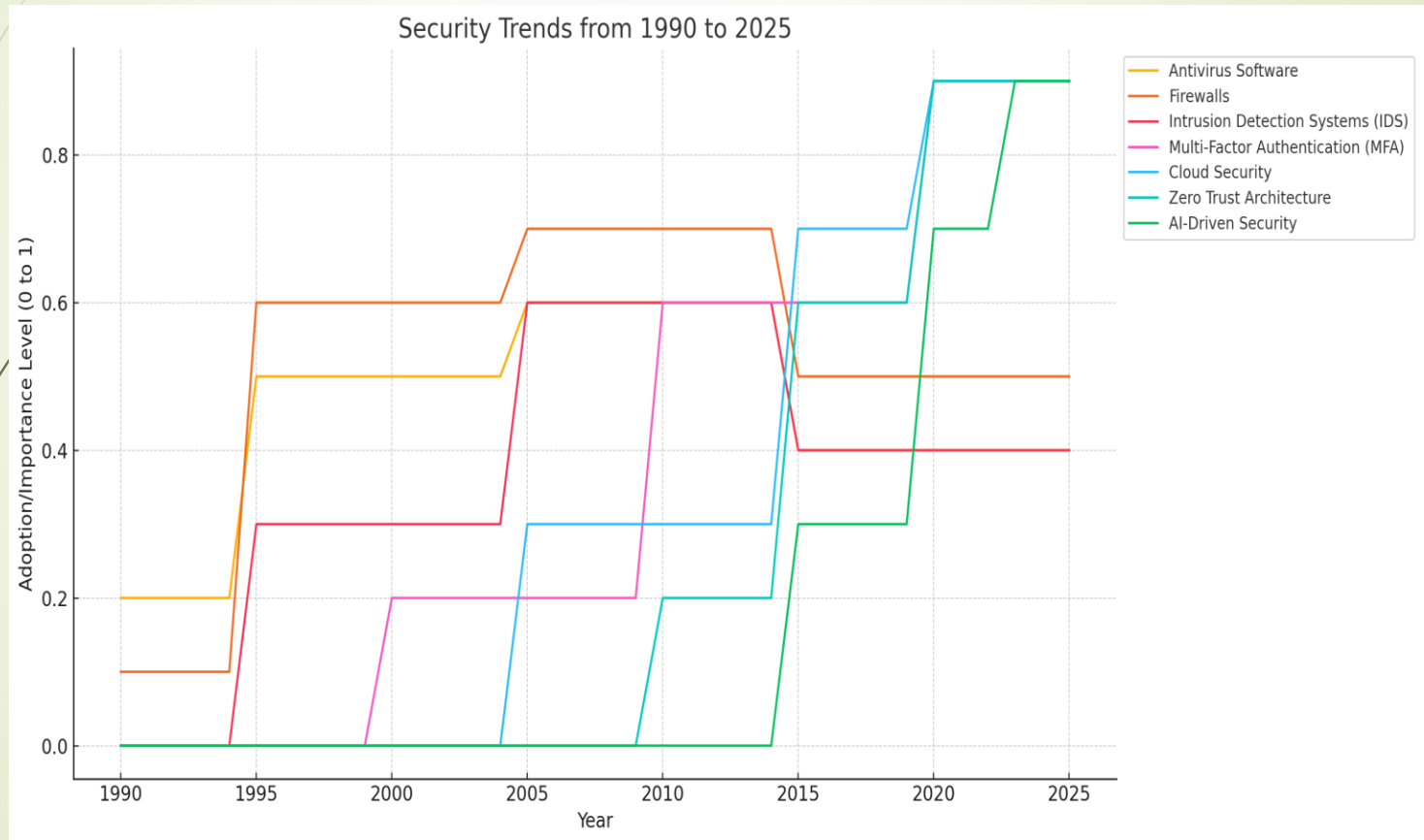
- A newer area, focusing on protecting against the misuse of generative AI tools and securing AI systems.



Aim of Course

- ▶ our focus is on Computer, Mobile, Wireless, and Cloud Security
 - ▶ which consists of measures to deter, prevent, detect, and correct security violations that involve the transmission & storage of information
- 

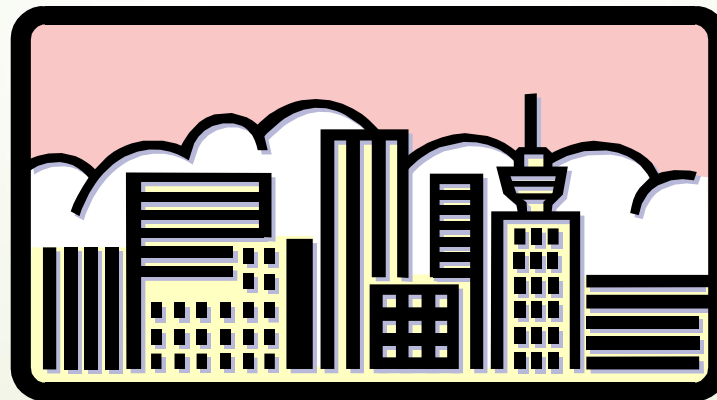
Security Trends



Zero Trust Architecture: **Never Trust, Always Verify:** This core principle dictates that no user, device, or application is trusted by default. Every access request must be authenticated and authorized based on established policies.

Security Architecture

- defines a systematic way of defining and providing security requirements
- for us it provides a useful, if abstract, overview of concepts we will study





Aspects of Security

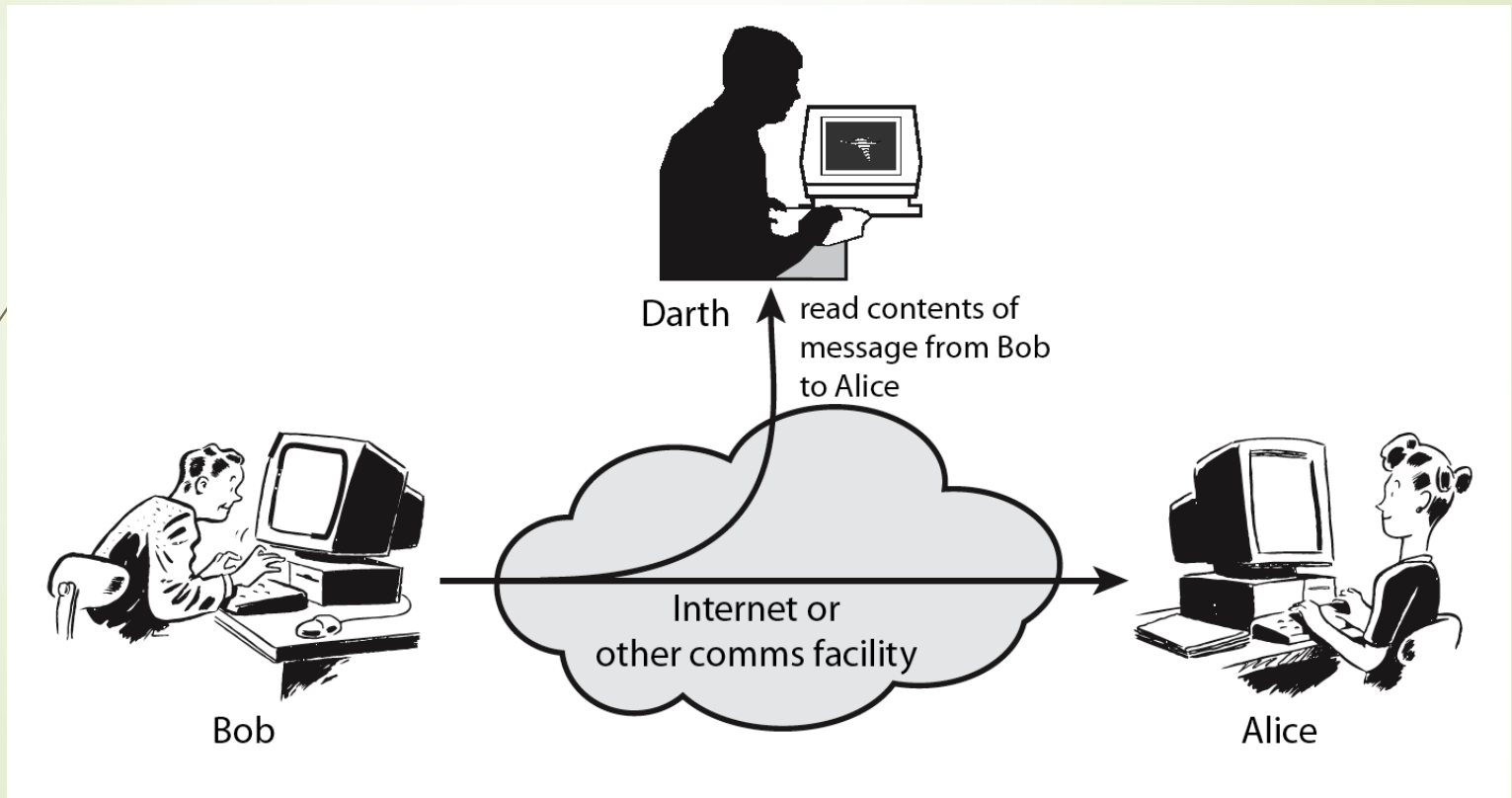
- consider 3 aspects of information security:
 - **security attack**
 - **security mechanism**
 - **security service**



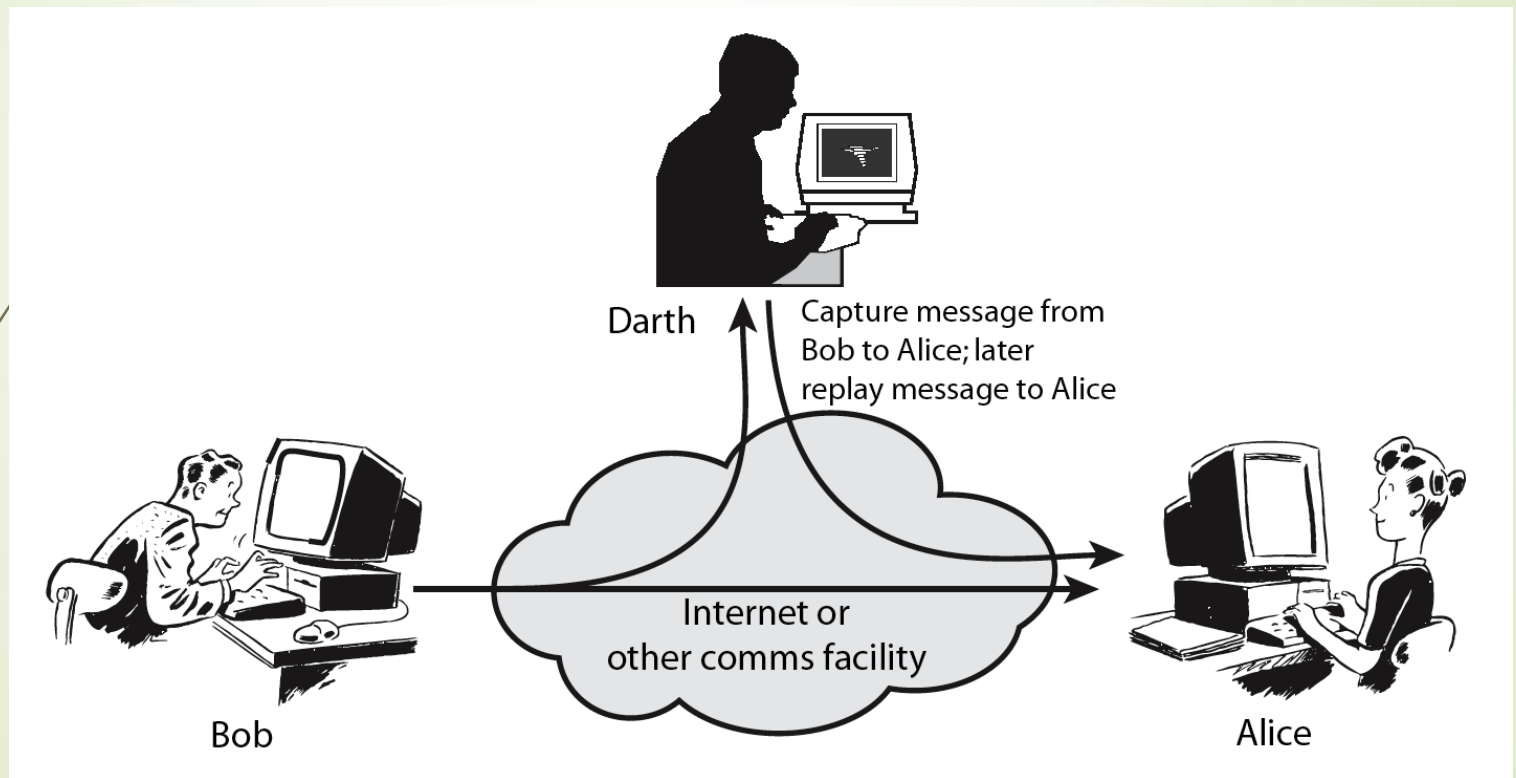
Security Attack

- any action that compromises the security of information owned by an organization
- information security is about how to prevent attacks, or failing that, to detect attacks on information-based systems
- often *threat* & *attack* used to mean same thing
- have a wide range of attacks
- can focus of generic types of attacks
 - passive
 - active

Passive Attacks



Active Attacks





Difference between Computer Security and Cybersecurity

➤ Computer Security:

- **Focus:** Protecting individual computer systems and the data stored on them.
- **Scope:** Includes measures like hardware and software security, access control, and data encryption within a specific computer or set of computers.
- **Examples:** Antivirus software, firewalls, password protection, and physical security measures for computers.

➤ Cybersecurity:

- **Focus:** Protecting digital systems, networks, and data from cyber threats.
- **Scope:** Broader than computer security, encompassing the security of entire networks, online systems, and digital information.
- **Examples:** Network security, cloud security, web application security, incident response, and risk management.



Overview of Computer Security

- **Computer security** is the practice of protecting computers, networks, and data from unauthorized access, attacks, and damage.
- It involves:
 - Protecting **hardware** (computers, routers).
 - Securing **software** (operating systems, apps).
 - Safeguarding **data** (personal and organizational).

Importance of Cybersecurity

- **Cybersecurity** is the practice of protecting internet-connected systems, networks, and data from digital attacks. It involves using various technologies, processes, and policies to safeguard against threats like malware, ransomware, phishing, and data breaches. Effective cybersecurity is crucial for individuals and organizations to maintain confidentiality, integrity, and availability of information and systems.
 - We use computers for **banking, communication, education, and healthcare**.
 - Cyber attacks can lead to **data theft, financial loss, and privacy breaches**.
 - Organizations need cybersecurity to protect **customer data and their reputation**.
 - It ensures **business continuity** and compliance with legal requirements.
 - **Example:**
A hospital's system without security could allow attackers to access patient medical records.



Threat Landscape

- The **threat landscape** refers to the different types of cyber threats faced today, including:
- **Malware:** Malicious software like viruses, worms, ransomware.
- **Phishing:** Fake emails to steal login information.
- **Denial-of-Service (DoS) Attacks:** Overloading systems to make them unavailable.
- **Insider threats:** Employees misusing access.
- **Zero-day attacks:** Exploiting software vulnerabilities before they are patched.
- Effective protection against the threat landscape requires a comprehensive approach to cybersecurity. Organizations must adopt proactive measures to mitigate potential threats and reduce vulnerabilities.



CIA Triad: Confidentiality, Integrity, Availability

- The **CIA Triad** is the foundation of cybersecurity:
- **Confidentiality:** Ensuring that information is accessible only to those authorized.
 - *Example:* Using passwords and encryption to protect data.
- **Integrity:** Ensuring data is accurate and unaltered.
 - *Example:* Using checksums to detect data tampering.
- **Availability:** Ensuring authorized users can access data when needed.
 - *Example:* Using backups and redundant systems to avoid downtime.



Fundamentals of Cryptography

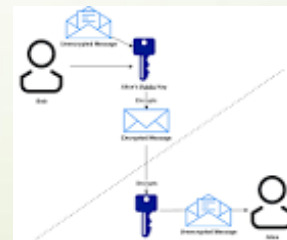
- **Cryptography** is the science of securing communication and data using codes so that only authorized parties can read it.
- Used to:
- Protect **confidentiality** of data.
- Ensure **integrity** (data has not been changed).
- Enable **authentication** (confirm identity).
- Support **non-repudiation** (sender cannot deny sending).

History of Cryptography

- **Ancient times:** Caesar cipher used by Julius Caesar to shift letters in messages.
- **World War II:** Enigma machine used by Germany, broken by Allied cryptographers.



- **Modern cryptography:** Uses complex mathematics and computers for secure encryption.





Encryption and Decryption

➤ Encryption:

- To ensure the confidentiality and security of data by making it unreadable to anyone who doesn't possess the correct decryption key.
- Involves using an encryption algorithm and a key to transform plaintext into ciphertext.
- Imagine locking a door with a key. The lock is the encryption, and the key is the decryption key. Only someone with the key can unlock the door and access the contents inside.

➤ Decryption:

- To make encrypted information readable again.
- Uses a decryption key (which may be the same as the encryption key in symmetric encryption or a different, related key in asymmetric encryption) to reverse the encryption process and convert ciphertext back into plaintext.
- Using the key to unlock the door and gain access to the contents inside.



Symmetric vs. Asymmetric Encryption

➤ ◆ Symmetric Encryption

- Uses the **same key for encryption and decryption**.
- Fast and efficient.
- Key distribution can be challenging.
- *Example: **AES (Advanced Encryption Standard)***

➤ ◆ Asymmetric Encryption

- Uses **two keys**: a public key (for encryption) and a private key (for decryption).
- Solves key distribution problem.
- Slower than symmetric encryption.
- *Example: **RSA (Rivest-Shamir-Adleman)***



Common Algorithms (AES, RSA)

➤ **AES (Advanced Encryption Standard):**

- Symmetric encryption algorithm.
- Block cipher with 128, 192, or 256-bit keys.
- Widely used for secure data encryption in government and industry.

➤ **RSA:**

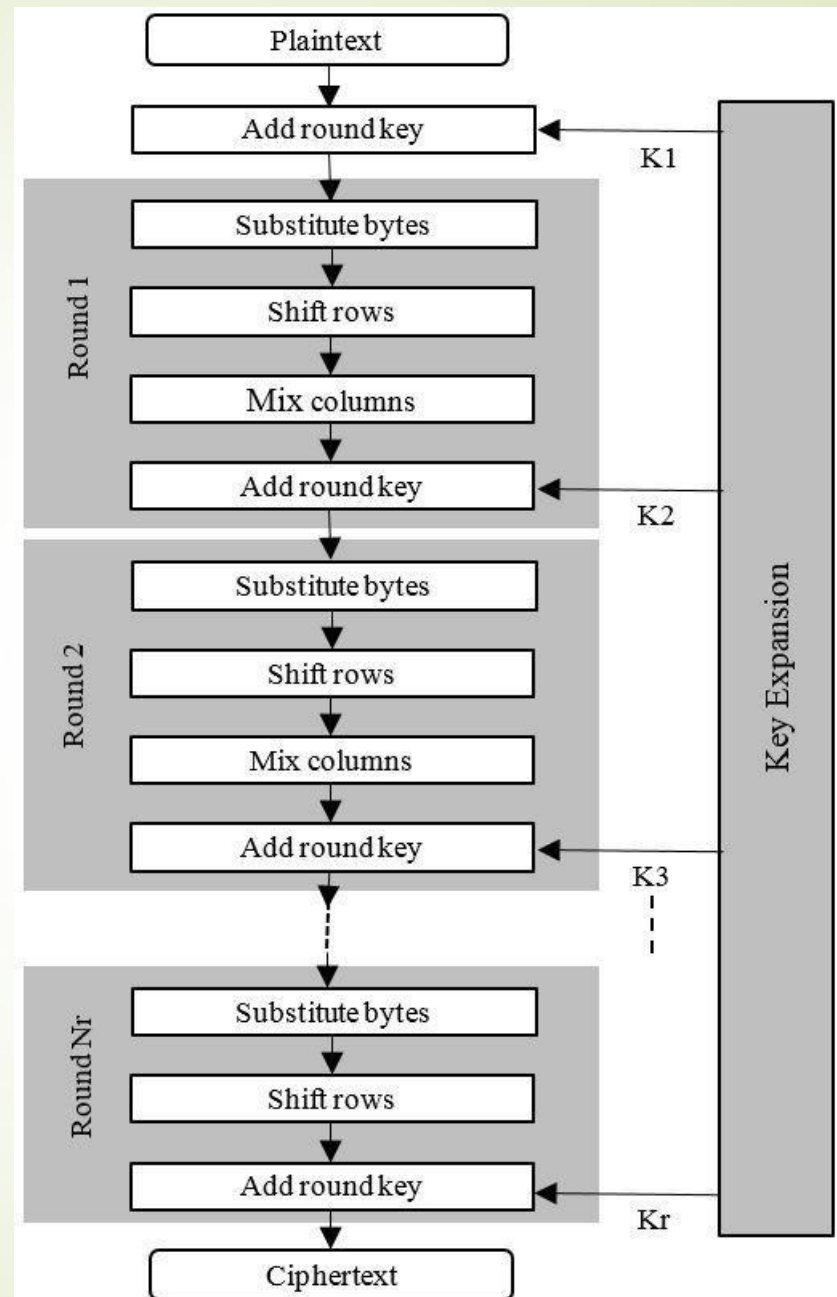
- Asymmetric encryption algorithm.
- Based on the mathematical difficulty of factoring large prime numbers.
- Used for secure data transmission and digital signatures.



Advanced Encryption Standard (AES)

- **What is AES?**
- AES is a **symmetric key encryption algorithm**, meaning **the same key is used for both encryption and decryption**.
- It is the **current encryption standard used by governments, banks, and industry** for secure data protection.

How AES Works:



How AES Works:

➤ Block Cipher:

- Encrypts **fixed-size blocks of data (128 bits)** at a time.
- Uses **key sizes of 128, 192, or 256 bits** (AES-128, AES-192, AES-256).

➤ Rounds:

- Performs **multiple rounds of transformation** on data:
 - AES-128: 10 rounds
 - AES-192: 12 rounds
 - AES-256: 14 rounds

➤ Main Operations in Each Round:

- **SubBytes:** Substitute each byte using a substitution box (S-box) for non-linearity.
- **ShiftRows:** Rows of the matrix are shifted to add diffusion.
- **MixColumns:** Columns are mixed to further obscure data.
- **AddRoundKey:** A portion of the encryption key is added using XOR.

➤ Key Expansion:

- AES expands the original key into multiple **round keys** used for each round.

AES Transformation functions

➤ 1. SubBytes (Byte Substitution):

- This step substitutes each byte of the state with another byte, using a substitution box (S-box). The S-box is a pre-defined lookup table that provides a non-linear transformation, increasing the algorithm's complexity.

➤ 2. ShiftRows:

- This transformation shifts the bytes in each row of the state array cyclically to the left. The first row remains unchanged, the second row is shifted by one position, the third by two, and the fourth by three. This step helps to ensure that bytes in each column are not encrypted independently.

➤ 3. MixColumns:

- This transformation operates on each column of the state, using a mathematical function to mix the bytes within the column. This mixing process ensures that changes in one byte of the state will affect multiple bytes in the next round.

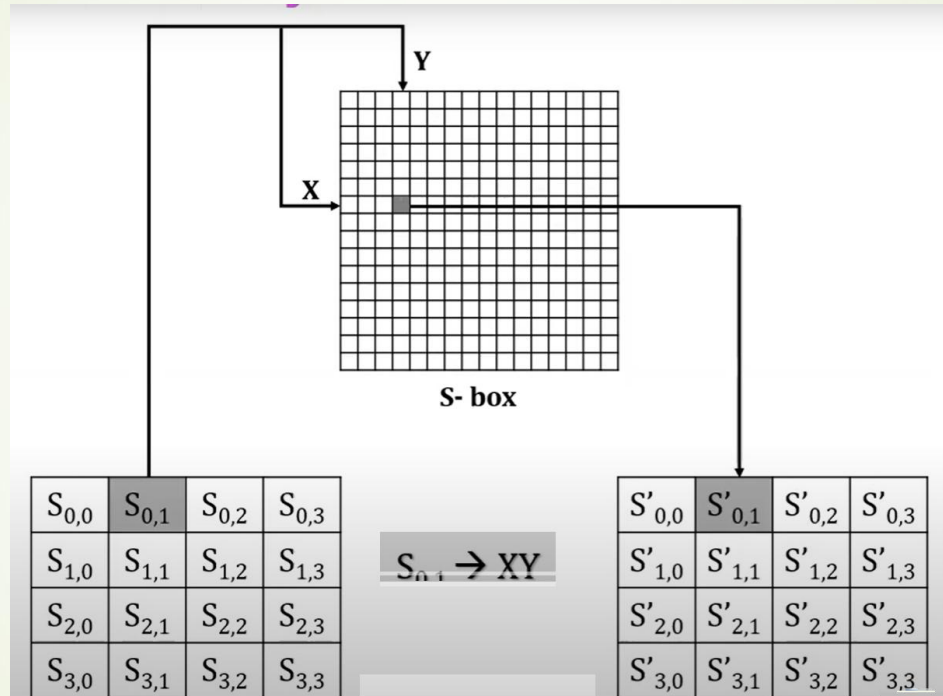
➤ 4. AddRoundKey:

- In this step, a round key, derived from the main encryption key, is XORed with the current state. This process adds a layer of key-dependent randomness to the encryption process.

1. AES substitution box (S-box)

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
00	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
10	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
20	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
30	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
40	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
50	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
60	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
70	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
80	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
90	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A0	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B0	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C0	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D0	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E0	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F0	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

1. SubBytes (Byte Substitution)



Byte Substitution Example

83 → EC

EA	04	65	85
83	45	5D	96
5C	33	98	B0
F0	2D	AD	C5



87	F2	4D	97
?			

2. ShiftRows:

- the current State Matrix is

$$\begin{pmatrix} 63 & EB & 9F & A0 \\ C0 & 2F & 93 & 92 \\ AB & 30 & AF & C7 \\ 20 & CB & 2B & A2 \end{pmatrix}$$

- four rows are shifted cyclically to the left by offsets of 0,1,2, and 3
- the new State Matrix is

$$\begin{pmatrix} 63 & EB & 9F & A0 \\ 2F & 93 & 92 & C0 \\ AF & C7 & AB & 30 \\ A2 & 20 & CB & 2B \end{pmatrix}$$

- this linear mixing step causes diffusion of the bits over multiple rounds

3. MixColumns:

- **State Matrix Representation** – The data being processed is stored in a 4x4 grid called the state matrix.
- **Fixed Matrix** – A fixed matrix is used in the MixColumns transformation. This matrix remains same throughout the encryption and decryption procedure.
- **Multiplication Column-wise** – Each column of the state matrix is considered a polynomial in $GF(2^8)$. The polynomial is then multiplied by the fixed matrix's corresponding column.
- **Polynomial Multiplication** – When multiplying polynomials in $GF(2^8)$, there are a few rules that must be followed. Firstly, modulo arithmetic must be used with a fixed polynomial called the irreducible polynomial. In the case of AES, this polynomial is $x^8 + x^4 + x^3 + x + 1$ represented in hexadecimal as 0x11B.
- **XOR Operation** – Following the multiplication stage, the results are XORed (exclusively ORed) to produce the final result for each column.
- **Resultant State Matrix** – By executing the multiplication and XOR operations for every column, the converted state matrix—which shows the MixColumns transformation—is produced.

MixColumns example

Let's take this example:

The diagram illustrates the MixColumns operation. On the left, a column vector of four bytes is labeled $a_0 - a_3$. These bytes are d4, bf, 5d, and 30. This vector is multiplied (indicated by a dot) by a 4x4 matrix. The matrix contains the following values:

02	03	01	01
01	02	03	01
01	01	02	03
03	01	01	02

The result of the multiplication is a new column vector on the right, labeled $r_0 - r_3$. The bytes in this vector are 04, 66, 81, and e5.

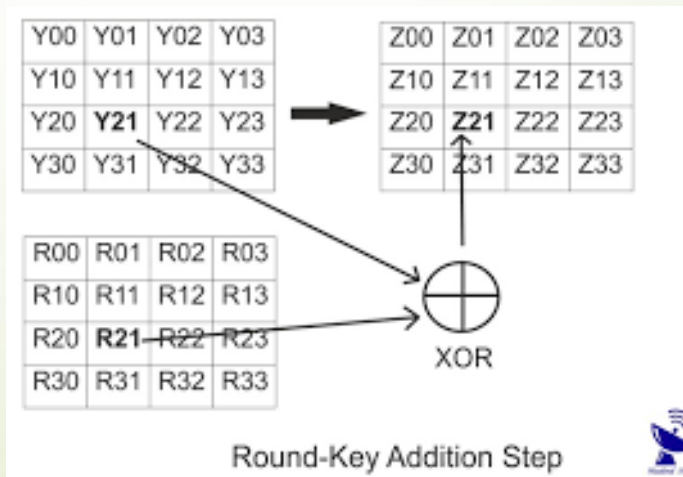
$$r_0 = \{02.d4\} + \{03.bf\} + \{01.5d\} + \{01.30\}$$

4. AddRoundKey:

•**Key Addition** – Each byte of the data is combined with a corresponding byte of the encryption key using a simple bitwise XOR operation in the AddRoundKey transformation. It means that if the bit in the data and the corresponding bit in the key disagree, the resulting bit in the encrypted data will be set to 1; otherwise, it will be set to 0.

•**Key Expansion** – This process encrypts the encryption key before applying the AddRoundKey transformation. This process creates a unique set of round keys for each encryption round.

•**Round Keys** – Each round key is created from the original encryption key and is used to encrypt a particular round of data. The AddRoundKey transformation combines each byte of the input with the matching byte of the round key to produce the encrypted output.



AES Encrypt and Decrypt Pseudocode

```
def encrypt_block(block, round_keys):
    state = [list(block[i:i+4]) for i in range(0, 16, 4)]
    add_round_key(state, round_keys[0])
    for rnd in range(1, 10):
        sub_bytes(state)
        shift_rows(state)
        mix_columns(state)
        add_round_key(state, round_keys[rnd])
    sub_bytes(state)
    shift_rows(state)
    add_round_key(state, round_keys[10])
    return bytes(sum(state, []))

def decrypt_block(block, round_keys):
    state = [list(block[i:i+4]) for i in range(0, 16, 4)]
    add_round_key(state, round_keys[10])
    for rnd in range(9, 0, -1):
        inv_shift_rows(state)
        inv_sub_bytes(state)
        add_round_key(state, round_keys[rnd])
        inv_mix_columns(state)
    inv_shift_rows(state)
    inv_sub_bytes(state)
    add_round_key(state, round_keys[0])
    return bytes(sum(state, []))
```



AES

- Example Scenario:
 - Encrypting a file before uploading to cloud storage. AES is used in SSL/TLS for secure web browsing.
- Advantages of AES:
 - Fast and efficient for large volumes of data. Strong security against brute-force attacks. Lightweight enough for use in mobile and embedded systems.



Applications of AES:

- AES is used in a wide range of applications, including:
 - Wireless security: Securing Wi-Fi networks.
- Encrypted Browsing: Securing websites using SSL/TLS protocols.
- File encryption: Protecting sensitive files like documents, images, and chat messages.
- Processor security: Hardware-level encryption in some processors.
- Data storage: Securing data stored on hard drives, cloud servers, and other storage media.



RSA (Rivest-Shamir-Adleman)

- **What is RSA?**
- RSA is an **asymmetric encryption algorithm**, meaning it uses **two keys**:
 - **Public key**: Used to encrypt data.
 - **Private key**: Used to decrypt data.
- It is widely used for **secure data transmission and digital signatures**.

How RSA Works:

➤ Key Generation:

1. Choose two large **prime numbers** (p and q).

2. Compute:

$$n = p \times q$$

3. Compute **Euler's totient**:

$$\phi(n) = (p - 1) \times (q - 1)$$

4. Choose an encryption exponent e such that:

$$1 < e < \phi(n)$$

and e is co-prime to $\phi(n)$.

5. Compute decryption exponent d :

$$d \equiv e^{-1} \pmod{\phi(n)}$$

How RSA Works:

➤ Keys: Public key: (e,n) Private key: (d,n)

➤ Encryption:

If m is the plaintext (converted to a number),

$$c = m^e \mod n$$

➤ Decryption:

$$m = c^d \mod n$$

➤ Recovers the original message m .

RSA (Rivest-Shamir-Adleman)

– Pseudocode

```
# RSA key generation
def generate_keys(bits=512):
    p = generate_large_prime(bits)
    q = generate_large_prime(bits)
    while q == p:
        q = generate_large_prime(bits)
    n = p * q
    phi = (p - 1) * (q - 1)
    e = 65537
    if gcd(e, phi) != 1:
        e = 3
        while gcd(e, phi) != 1:
            e += 2
    d = modinv(e, phi)
    return ((e, n), (d, n))

# Encryption:  $c = m^e \bmod n$ 
def encrypt(message, public_key):
    e, n = public_key
    message_bytes = message.encode()
    message_int = int.from_bytes(message_bytes, byteorder='big')
    if message_int >= n:
        raise ValueError("Message too large for the key size. Use a larger key or split the message.")
    cipher_int = pow(message_int, e, n)
    return cipher_int

# Decryption:  $m = c^d \bmod n$ 
def decrypt(cipher_int, private_key):
    d, n = private_key
    message_int = pow(cipher_int, d, n)
    message_length = (message_int.bit_length() + 7) // 8
    message_bytes = message_int.to_bytes(message_length, byteorder='big')
    return message_bytes.decode()
```




RSA

- Example Scenario:

- Encrypting data during secure email transmission (PGP).SSL/TLS certificate authentication.

- Advantages of RSA:

- Solves the key distribution problem (public key can be shared openly).Used for secure key exchange in combination with symmetric algorithms (RSA exchanges the AES key securely).



Summary



- Computer security protects computers and data from unauthorized access, damage, and disruption.
- It is crucial for protecting personal data, organizational assets, financial systems, and national security infrastructure.
- Evolving threats require constant monitoring, updated security practices, and user awareness to mitigate risks.
- Cryptography is the science of securing information by transforming it into unreadable formats for unauthorized users.